

Tarea sección 2.7

5) $y = 4x - 3x^2, \quad (2, -4)$

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow 2} \frac{4x - 3x^2 - (-4)}{x - 2}$$

$$m = \lim_{x \rightarrow 2} \frac{-3x^2 + 4x + 4}{x - 2} = \lim_{x \rightarrow 2} \frac{-(x-2)(3x+2)}{x-2} = -3(2) + 2 = -8$$

$$y - (-4) = -8(x - 2)$$

$$y = -8x + 12$$

7) $y = \sqrt{x} \quad (1, 1)$

$$m = \lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1} = \frac{0}{0}$$

$$\lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1} = \frac{\sqrt{x} - 1}{(\sqrt{x} - 1)(\sqrt{x} + 1)} = \frac{1}{\sqrt{x} + 1} = \frac{1}{1 + 1} = \frac{1}{2}$$

$$y - 1 = \frac{1}{2}(x - 1)$$

$$y = \frac{1}{2}x - \frac{1}{2} + 1$$

$$y = \frac{1}{2}x + \frac{1}{2}$$

73) $y = 10t - 4.9t^2$

$$t = 2$$

$$10 \text{ m/s}$$

$$V(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{10(a+h) - 4.9(a+h)^2 - (10a - 4.9a^2)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{10a + 10h - 4.9(a^2 + 2ah + h^2) - 10a + 4.9a^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{10h - 4.9a^2 - 9.8ah - 4.9h^2 + 4.9a^2}{h} = \lim_{h \rightarrow 0} (10 - 9.8a - 4.9h)$$

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$$\lim_{h \rightarrow 0} 70 - 9.8a - 4.9h$$

$$V(a) = 70 - 9.8a \quad V(2) = 70 - 9.8(2) = -9.6 \text{ m/s}$$

17) El orden en forma creciente es $= g'(0), 0, g'(4), g'(2)$
y $g'(-2)$

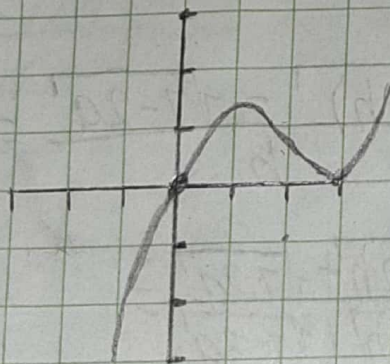
18) a) Calcule el valor $f'(50)$

$$f'(50) = 26$$

b) ¿Es $f'(10) > f'(30)$ = No

c) ¿Es $f'(60) > \frac{f(80) - f(40)}{80 - 40}$ = Si

23) Trace la gráfica; $f(0) = 0, f'(0) = 3, f'(7) = 0, y f'(2) = -1$



$$4) f(x) = 3x^2 - x^3 \quad f'(1) = ? \quad (7, 2)$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{3(x+h)^2 - (x+h)^3 - (3x^2 - x^3)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3(x^2 + 2xh + h^2) - (x^3 + 3x^2h + 3xh^2 + h^3) - 3x^2 + x^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{3x^2} + 6xh + 3h^2 - \cancel{x^3} - 3x^2h - 3xh^2 - h^3 - \cancel{3x^2} + \cancel{x^3}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 3x^2h - 3xh^2 - h^3 + x^3}{h}$$

$$\lim_{h \rightarrow 0} \frac{h(6x + 3h - 3x^2 - 3xh - h^2)}{h} = \lim_{h \rightarrow 0} (6x + 3h - 3x^2 - 3xh - h^2)$$

$$f'(x) = 6x - 3x^2 \quad f'(1) = 6(1) - 3(1)^2 = 3$$

$$y - 2 = 3(x - 1)$$

$$y = 3x - 7$$

$$35) f(x) = \sqrt{7 - 2x}$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{7 - 2(a+h)} - \sqrt{7 - 2a}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{7 - 2a - 2h} - \sqrt{7 - 2a}}{h} \cdot \frac{\sqrt{7 - 2a - 2h} + \sqrt{7 - 2a}}{\sqrt{7 - 2a - 2h} + \sqrt{7 - 2a}} =$$

$$= \lim_{h \rightarrow 0} \frac{7 - 2a - 2h - (7 - 2a)}{h(\sqrt{7 - 2a - 2h} + \sqrt{7 - 2a})} = \frac{7 - 2a - 2h - 7 + 2a}{h(\sqrt{7 - 2a - 2h} + \sqrt{7 - 2a})} = \frac{-2}{\sqrt{7 - 2a - 2h} + \sqrt{7 - 2a}}$$

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$$= \frac{-2}{\sqrt{1-2a} + \sqrt{1-2a}} = \frac{-2}{1-2a}$$

$$39) \lim_{x \rightarrow 2} \frac{x^6 - 64}{x - 2}$$

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f(x) = x^6$$

$$a = 2$$

$$43) f(t) = 80t - 6t^2$$

$$v(a) = \lim_{h \rightarrow 0} \frac{80(a+h) - 6(a+h)^2 - (80a - 6a^2)}{h}$$

$$\lim_{h \rightarrow 0} \frac{80a + 80h - 6(a^2 + 2ah + h^2) - 80a + 6a^2}{h} = \frac{80h - 12ah - h^2}{h}$$

$$\lim_{h \rightarrow 0} (80 - 12a - h) = 80 - 12a$$

$$v(a) = 80 - 12a$$

$$v(4) = 80 - 12(4) = 32 \text{ m/s}$$

